WORKED SOLUTION

STEP II 2000, Question 5

Pure • Least Squares Approximation

STEP II 2000, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q5

The key idea

Once you expand the square, the quantity to be minimised is just a quadratic in λ, with coefficients that are fixed numbers (integrals involving f). Completing the square gives the best λ and the smallest error in one go. For the sine example, the point is that $\sin(\pi x/n)$ is almost a straight line through the origin when n is large, and the small-angle approximations make this precise.

Solution

The best value of λ

Write

$$J = \int_0^1 xf(x)\,\mathrm{d}x, \qquad K = \int_0^1 (f(x))^2\,\mathrm{d}x.$$

These are constants, not depending on λ. Expanding the square and using $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$,

$$\begin{aligned} I(\lambda) = \int_0^1 (f(x) - \lambda x)^2\,\mathrm{d}x &= K - 2\lambda J + \tfrac{1}{3}\lambda^2 \\ &= \tfrac{1}{3}(\lambda - 3J)^2 + K - 3J^2. \end{aligned}$$

The squared term is never negative and is zero only when $\lambda = 3J$, so $I(\lambda)$ is smallest exactly when

$$\lambda = 3\int_0^1 xf(x)\,\mathrm{d}x,$$

as required. (Equivalently, $I'(\lambda) = -2J + \frac{2}{3}\lambda = 0$ gives $\lambda = 3J$, and $I''(\lambda) = \frac{2}{3} > 0$ confirms a minimum.)

The residual error

R^2 is the minimum value of I, which the completed square shows is $K - 3J^2$. Since $J = \lambda/3$, we have $3J^2 = \frac{1}{3}\lambda^2$, so

$$R^2 = \int_0^1 (f(x))^2 \,\mathrm{d}x - \tfrac{1}{3}\lambda^2,$$

as required. (If you found λ by differentiating instead, substitute $J = \lambda/3$ into $K - 2\lambda J + \frac{1}{3}\lambda^2$ to get $K - \frac{2}{3}\lambda^2 + \frac{1}{3}\lambda^2$, the same thing.)

The case $f(x) = \sin(\pi x/n)$

To keep the algebra tidy, put $a = \pi/n$, so $f(x) = \sin ax$ and $a \to 0$ as $n \to \infty$. Integrating by parts with $u = x$ and $\mathrm{d}v = \sin ax\,\mathrm{d}x$,

$$\begin{aligned}\int_0^1 x \sin ax \,\mathrm{d}x &= \Big[-\frac{x\cos ax}{a}\Big]_0^1 + \frac{1}{a}\int_0^1 \cos ax \,\mathrm{d}x \\ &= -\frac{\cos a}{a} + \frac{\sin a}{a^2},\end{aligned}$$

so

$$\lambda = \frac{3(\sin a - a\cos a)}{a^2}.$$

Ask yourself Why do we need the θ^3 term in the approximation to $\sin\theta$, rather than just $\sin\theta \approx \theta$?

With only $\sin a \approx a$ and $\cos a \approx 1$, the numerator becomes $a - a = 0$, which tells us nothing: the leading terms cancel, and the answer lives in the next order. Using both approximations from the hint,

$$\sin a - a\cos a \approx \left(a - \tfrac{1}{6}a^3\right) - a\left(1 - \tfrac{1}{2}a^2\right) = \tfrac{1}{3}a^3,$$

and therefore

$$\lambda \approx \frac{3\cdot\frac{1}{3}a^3}{a^2} = a = \frac{\pi}{n}$$

for large n. This is result (i).

For (ii) we also need K. Using $\sin^2 ax = \frac{1}{2}(1 - \cos 2ax)$,

$$K = \int_0^1 \sin^2 ax \,\mathrm{d}x = \frac{1}{2} - \frac{\sin 2a}{4a}.$$

As $n \to \infty$, $a \to 0$ and $\dfrac{\sin 2a}{2a} \to 1$, so $K \to \frac{1}{2} - \frac{1}{2} = 0$. Also $\lambda \to 0$, since $\lambda \approx \pi/n$. Hence

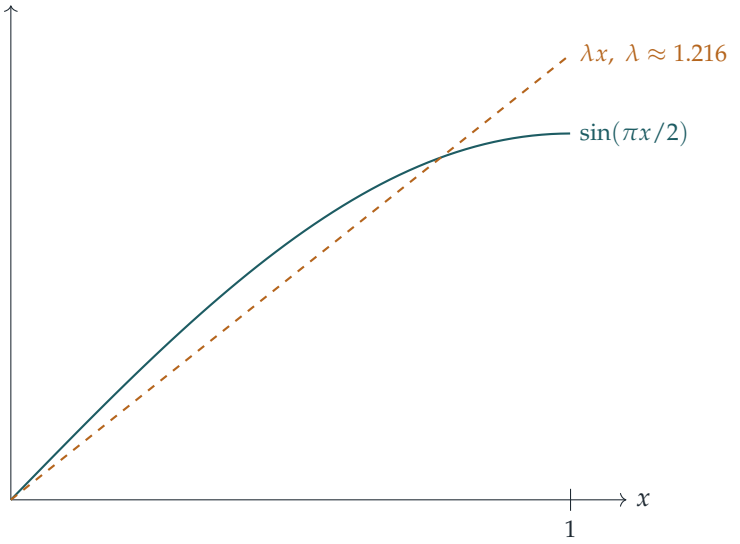
$$R^2 = K - \tfrac{1}{3}\lambda^2 \to 0 - 0 = 0,$$

and because $R \geq 0$ we get $\lim_{n\to\infty} R = 0$. This is result (ii).

In fact the cancellation is much stronger than this. With $\sin 2a \approx 2a - \frac{4}{3}a^3$ we get $K \approx \frac{1}{2} - (\frac{1}{2} - \frac{1}{3}a^2) = \frac{1}{3}a^2$, and $\frac{1}{3}\lambda^2 \approx \frac{1}{3}a^2$ too. So R^2 is not just small: it is small compared with K, the size of f itself. The straight line captures almost all of f.

Why these results are expected

On $0 \leq x \leq 1$ the angle $\pi x/n$ lies between 0 and π/n, which is small when n is large. So $\sin(\pi x/n) \approx \pi x/n$ throughout the interval: the function is itself very nearly the linear function λx with $\lambda = \pi/n$. The best-fitting line should therefore have gradient close to π/n, and the fit should become essentially perfect, so the residual error should tend to zero. The picture shows that even for $n = 2$ the curve is not far from a straight line.



Checks

- If $f(x) = cx$ already, then $\lambda = 3c\int_0^1 x^2\,\mathrm{d}x = c$ and $R^2 = \frac{1}{3}c^2 - \frac{1}{3}c^2 = 0$, as it should be.
- Numerically, for $n = 10$ the exact formula gives $\lambda = 0.31107$, against $\pi/10 = 0.31416$; for $n = 100$ the ratio $\lambda/(\pi/n)$ is 0.99990.
- For $n = 10$, quadrature gives $R^2 \approx 6.0 \times 10^{-7}$ while $K \approx 0.032$, so the residual really is tiny compared with f.

Common slips

- Stopping at $\sin a \approx a$, $\cos a \approx 1$ and getting $\lambda \approx 0$: true, but useless. Keep the cubic terms.
- Forgetting that $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$ is where the 3 in $\lambda = 3\int_0^1 xf(x)\,\mathrm{d}x$ comes from.

Beyond STEP

This is least-squares approximation, and it is really geometry. Treat functions on $[0,1]$ as vectors with "dot product" $\langle f, g\rangle = \int_0^1 fg\,\mathrm{d}x$. Then λx is the orthogonal projection of f onto the line spanned by x, and $R^2 = K - \frac{1}{3}\lambda^2$ is Pythagoras' theorem. Pushing the expansion one step further shows $R^2 \approx a^6/1575$, which comes from projecting the x^3 term away from x. Strang's *Introduction to Linear Algebra* develops projections and least squares, and the same ideas lead on to Fourier series.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.